ECE 313: Probability with Engineering Applications

2025 Fall   Instructors: Piao Chen & Xu Chen

Homework 8

Name: ______________________

Student ID: ______________________

Due Date: November 28 23:59, 2025

Problem 1. Of the voters in Illinois, a proportion p will vote for candidate A, and a proportion $1-p$ will vote for candidate B. In an election poll a number of voters are asked for whom they will vote. Let X_i be the indicator random variable with the value 1 if the ith person interviewed will vote for A, and 0 for B. A model for the election poll is that the persons interviewed are chosen in a random way such that the indicator random variables $X_1, X_2, \ldots$ are independent and have a Ber(p) distribution.

(a) Suppose we use $\bar{X}_n$ to predict p. According to Chebyshev's inequality, how large should n be chosen so that the probability that $\bar{X}_n$ is within 0.2 of the "true" p is at least 0.8?

(b) If $p > \frac{1}{2}$ then candidate G wins; if $\bar{X}_n > \frac{1}{2}$ you predict that G will win. Find an n (as small as you can) so that the probability that your prediction is correct is at least 0.8, assuming that in fact $p = 0.6$.

Problem 2. Let the random variable X be uniformly distributed on the interval $[0, 1]$. Define

$$g(x) = \frac{1}{1+x}, \qquad f(x) = e^x.$$

(a) Determine whether $f(x)$ and $g(x)$ are convex or concave on the interval $[0, 1]$. (You may base your answer on derivatives or the graph of the function; no formal proof of the definition is required.)

(b) Using Jensen's inequality, provide an appropriate upper or lower bound for the quantities

$$\mathbb{E}[g(X)] \quad \text{and} \quad \mathbb{E}[f(X)].$$

Problem 3. Let the joint pdf of X and Y be given by

$$f(x, y) = e^{-x/\alpha}\, y\, e^{-y^2}, \qquad \text{for } x > 0,\ y > 0,$$

where $\alpha \neq 0$. The random variables X and Y are said to have a two-dimensional (or bivariate) normal pdf.

(a) Show the marginal pdf's of X and Y

(b) Find the values of α for which X and Y are independent.

Problem 4. Suppose that the joint distribution function of X and Y is given by

$$F(x, y) = 1 - e^{-3x} - e^{-y} + e^{-(3x+y)}, \qquad \text{if } x > 0,\ y > 0,$$

and $F(x, y) = 0$ otherwise.

(a) Determine the marginal distribution functions of X and Y.

(b) Determine the joint probability density function of X and Y.

(c) Determine the marginal probability density functions of X and Y.

(d) Find out whether X and Y are independent.